

Milestone 2 Device Configuration Commands

This worksheet documents the final Milestone 2 Packet Tracer configuration. Cisco IOS commands are shown for routers and switches. Packet Tracer server services and server IP settings were configured through the server GUI, not Cisco CLI.

DLS1: Catalyst 3560 distribution switch

```
enable
configure terminal
hostname DLS1
no ip domain-lookup
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 1,10,20,30,40,50,99,100 root primary
vlan 10
  name ADMIN_HR
vlan 20
  name ENGINEERING_DESIGN
vlan 30
  name PRODUCTION_FLOOR
vlan 40
  name FINANCE_PROCURE
vlan 50
  name ICT_TECHNICAL
vlan 99
  name MGMT_SERVERS
vlan 100
  name GUEST_WIFI
vlan 999
  name UNUSED_NATIVE
exit
interface Vlan10
  ip address 192.168.35.33 255.255.255.240
  ip helper-address 192.168.35.114
  no shutdown
interface Vlan20
  ip address 192.168.35.1 255.255.255.224
  ip helper-address 192.168.35.114
  no shutdown
interface Vlan30
  ip address 192.168.35.49 255.255.255.240
  ip helper-address 192.168.35.114
  no shutdown
interface Vlan40
  ip address 192.168.35.97 255.255.255.240
  ip helper-address 192.168.35.114
  no shutdown
interface Vlan50
  ip address 192.168.35.65 255.255.255.240
  ip helper-address 192.168.35.114
  no shutdown
interface Vlan99
  ip address 192.168.35.113 255.255.255.240
  ip helper-address 192.168.35.114
  no shutdown
interface Vlan100
  ip address 192.168.35.81 255.255.255.240
  ip helper-address 192.168.35.114
  no shutdown
interface GigabitEthernet0/1
  no switchport
  ip address 192.168.35.254 255.255.255.252
  no shutdown
```

```

router ospf 1
  router-id 2.2.2.2
  passive-interface default
  no passive-interface GigabitEthernet0/1
  network 192.168.35.0 0.0.0.255 area 0
exit
ip route 0.0.0.0 0.0.0.0 192.168.35.253
interface range FastEthernet0/21 - 24
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport nonegotiate
  spanning-tree guard root
  no shutdown
exit
interface FastEthernet0/21
  switchport trunk allowed vlan 1,20,50,99
interface FastEthernet0/22
  switchport trunk allowed vlan 1,20,50,99
interface FastEthernet0/23
  switchport trunk native vlan 1
  switchport trunk allowed vlan 1,30,99,100
interface FastEthernet0/24
  switchport trunk native vlan 999
  switchport trunk allowed vlan 40,999
interface GigabitEthernet0/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 1,10,99
  no shutdown
interface FastEthernet0/2
  switchport mode access
  switchport access vlan 99
  spanning-tree portfast
  no shutdown
interface FastEthernet0/3
  switchport mode access
  switchport access vlan 99
  spanning-tree portfast
  no shutdown
end
copy running-config startup-config

```

R-EDGE: ISR 2911 core router

```

enable
configure terminal
hostname R-EDGE
no ip domain-lookup
interface GigabitEthernet0/0
  ip address dhcp
  ip nat outside
  no shutdown
interface GigabitEthernet0/1
  ip address 192.168.35.253 255.255.255.252
  ip nat inside
  no shutdown
router ospf 1
  router-id 1.1.1.1
  network 192.168.35.252 0.0.0.3 area 0
  default-information originate always
exit
ip route 0.0.0.0 0.0.0.0 dhcp

```

```
access-list 1 permit 192.168.35.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/0 overload
end
copy running-config startup-config
```

Access switch pattern

Use the matching VLAN and DLS1 uplink from the table below. The end-device port assignments follow the Milestone 1 logical topology.

Switch	User VLAN(s)	Access-switch uplink port(s)	DLS1 uplink port(s)	Primary allowed VLANs
ALS-ADM	10	Gi0/1	Gig0/2	1,10,99
ALS-ENG	20 and 50	Gi0/1 and Gi0/2	Fa0/21 and Fa0/22	1,20,50,99
ALS-PRD	30 and 100	Gi0/1	Fa0/23	1,30,99,100
ALS-FIN	40	Gi0/1	Fa0/24	40,999

For each access switch, configure the hostname, VLANs, user access ports, uplink, PortFast, and BPDU Guard. Keep unused ports shutdown.

ALS-ADM

```
enable
configure terminal
hostname ALS-ADM
vlan 10
  name ADMIN_HR
vlan 99
  name MGMT_SERVERS
interface range FastEthernet0/1 - 3
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation restrict
  switchport port-security mac-address sticky
  no shutdown
interface GigabitEthernet0/1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 1,10,99
  no shutdown
interface Vlan99
  ip address 192.168.35.115 255.255.255.240
  no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config
```

ALS-ENG

```
enable
configure terminal
hostname ALS-ENG
```

```

vlan 20
  name ENGINEERING_DESIGN
vlan 50
  name ICT_TECHNICAL
vlan 99
  name MGMT_SERVERS
interface range FastEthernet0/1 - 3
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation restrict
  switchport port-security mac-address sticky
  no shutdown
interface range FastEthernet0/17 - 19
  switchport mode access
  switchport access vlan 50
  spanning-tree portfast
  spanning-tree bpduguard enable
  no shutdown
interface GigabitEthernet0/1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 1,20,50,99
  no shutdown
interface GigabitEthernet0/2
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 1,20,50,99
  no shutdown
interface Vlan99
  ip address 192.168.35.116 255.255.255.240
  no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config

```

ALS-PRD

```

enable
configure terminal
hostname ALS-PRD
vlan 30
  name PRODUCTION_FLOOR
vlan 100
  name GUEST_WIFI
vlan 99
  name MGMT_SERVERS
interface range FastEthernet0/1 - 3
  switchport mode access
  switchport access vlan 30
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation restrict
  switchport port-security mac-address sticky
  no shutdown
interface GigabitEthernet0/1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk allowed vlan 1,30,99,100
  no shutdown

```

```

interface Vlan99
  ip address 192.168.35.117 255.255.255.240
  no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config

```

ALS-FIN

```

enable
configure terminal
hostname ALS-FIN
vlan 40
  name FINANCE_PROCURE
vlan 99
  name MGMT_SERVERS
vlan 999
  name UNUSED_NATIVE
interface range FastEthernet0/1 - 3
  switchport mode access
  switchport access vlan 40
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation shutdown
  switchport port-security mac-address sticky
  no shutdown
interface GigabitEthernet0/1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk native vlan 999
  switchport trunk allowed vlan 40,999
  no shutdown
interface Vlan99
  ip address 192.168.35.118 255.255.255.240
  no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config

```

Server and DHCP configuration

SVR-DC: DHCP + DNS server

SVR-DC is a Packet Tracer Server-PT device, so its IP settings and services were configured through the GUI rather than Cisco IOS CLI.

Setting	Value
IPv4 address	192.168.35.114
Subnet mask	255.255.255.240
Default gateway	192.168.35.113
DNS server	192.168.35.114
DHCP service	On
DNS service	On

DNS record	kagiso.local / file-server record to 192.168.35.119
------------	---

DHCP was configured in SVR-DC -> Services -> DHCP, not with IOS commands. The seven GUI scopes are:

Scope	Network/mask	Default gateway	DNS server
VLAN10_ADMIN	192.168.35.32/28	192.168.35.33	192.168.35.114
VLAN20_ENGINEERING	192.168.35.0/27	192.168.35.1	192.168.35.114
VLAN30_PRODUCTION	192.168.35.48/28	192.168.35.49	192.168.35.114
VLAN40_FINANCE	192.168.35.96/28	192.168.35.97	192.168.35.114
VLAN50_ICT	192.168.35.64/28	192.168.35.65	192.168.35.114
VLAN99_MANAGEMENT	192.168.35.112/28	192.168.35.113	192.168.35.114
VLAN100_GUEST	192.168.35.80/28	192.168.35.81	192.168.35.114

Evidence: 05_svr_dc_dhcp.png and 06_svr_dc_dns.png.

SVR-FP: file/application server on management VLAN

SVR-FP is also configured through the Packet Tracer GUI.

Setting	Value
IPv4 address	192.168.35.119
Subnet mask	255.255.255.240
Default gateway	192.168.35.113
HTTP service	On

Evidence: 07_svr_fp_http.png.

Final ACL policy (CR4)

The seven named inbound ACLs below are the final DLS1 policy. IOS sequence numbers are shown in their final corrected order. The file/application server is 192.168.35.119.

ACL-ENG-IN

```
10 deny ip any 192.168.35.96 0.0.0.15
20 permit ip any host 192.168.35.119
30 permit ip any any
```

Engineering traffic cannot reach Finance, but may reach the file server and other permitted destinations.

ACL-ADMIN-IN

```
10 deny ip any 192.168.35.96 0.0.0.15
20 permit ip any host 192.168.35.119
30 permit ip any any
```

Admin traffic cannot reach Finance, but may reach the file server and other permitted destinations.

ACL-FIN-IN

```
10 deny ip any 192.168.35.0 0.0.0.255
20 permit ip any host 192.168.35.119
30 permit ip any any
```

Finance is isolated from internal address space by the final applied policy.

ACL-PRD-IN

```
10 deny ip any host 192.168.35.119
20 deny ip any 192.168.35.96 0.0.0.15
30 permit ip any any
```

Production cannot reach the file server or Finance.

ACL-ICT-IN

```
10 deny ip any 192.168.35.96 0.0.0.15
20 permit ip any 192.168.35.112 0.0.0.15
30 permit ip any any
```

ICT cannot reach Finance and may reach the management/server subnet.

ACL-MGMT-IN

```
10 permit ip 192.168.35.64 0.0.0.15 any
20 permit ip 192.168.35.112 0.0.0.15 any
30 deny ip any any
```

Only ICT and the management/server subnet are permitted through the management policy.

ACL-GUEST-IN

```
10 deny ip any 192.168.35.0 0.0.0.255
20 permit ip any any
```

Guest traffic is denied access to internal RFC1918 project networks and permitted elsewhere.

The configured ACL names, entries, and counters are documented in [13_dls1_acls.png](#).

Verification commands

Run these before closing the M2 evidence pack:

```
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
show ip ospf neighbor
show spanning-tree root
show interfaces status
show access-lists
ping 192.168.35.119 source 192.168.35.1
ping 192.168.35.119 source 192.168.35.65
ping 192.168.35.119 source 192.168.35.97
nslookup file-server 192.168.35.114
```

Expected result: authorised Engineering / Admin / ICT users reach the server, while Finance and Production attempts are denied. Capture the successful ping, denied ping, DHCP lease output, and ACL hit counters as the final evidence set.

Final evidence checklist

- ☒ `show vlan brief` on DLS1 - 08_dls1_vlan_brief.png
- ☒ VLSM SVI addresses and DHCP helpers - 26_dls1_vlsm_svis_and_helpers.png
- ☒ `show interfaces trunk` on DLS1 and access switches - 09_dls1_trunks.png
- ☒ `show ip ospf neighbor` on R-EDGE and DLS1 - 11_dls1_ospf.png
- ☒ DHCP lease screenshots for VLAN 20 and VLAN 10 - 01_engineering_dhcp.png, 02_admin_dhcp.png
- ☒ DHCP lease screenshots for VLAN 100 and VLAN 40 - 08_guest_dhcp.png, 09_finance_dhcp.png
- ☒ ALS-ADM VLAN and trunk verification - 25_als_adm_vlan_and_trunk.png
- ☒ DHCP service screen on SVR-DC (GUI, not `show ip dhcp binding`) - 05_svr_dc_dhcp.png
- ☒ Successful ping from Engineering to 192.168.35.119 - 03_engineering_server_ping.png
- ☒ Failed ping from Production to 192.168.35.119 - 04_production_server_ping_denied.png
- ☒ `show access-lists` with deny counters incrementing - 13_dls1_acls.png
- ☒ STP root and redundant-link evidence - 15_stp_before_failover_part1.png, 16_stp_before_failover_part2.png, 17_stp_before_failover_part3.png, 18_stp_after_failover_part1.png, 22_stp_restored_part1.png
- ☒ Final Packet Tracer save for Milestone2/CLI066_Mbodi_40779750_M2_CONFIGURED.pkt